

AI-SA-IN-TH-000011

Module 8.1 · LLM Evaluation Fundamentals · AI Safety: Evaluation, Guardrails & Responsible Deployment

Bias Detection and Mitigation

Sources, Types, Detection Tools, and Mitigation Strategies for AI Bias

9 min

Duration

B4 – Analyse

Bloom's

L3 – Advanced

Level

Theory

Domain

Learning Objectives

- 1 Distinguish bias from hallucination and explain why the two require different mitigation strategies
- 2 Map the three sources of bias (data, algorithm, deployment) and identify failure points at each stage
- 3 Classify a given bias scenario into one of five bias types and select the appropriate detection method
- 4 Apply the 4/5ths disparate impact rule to a hiring dataset and determine whether bias is present
- 5 Compare pre-, in-, and post-processing mitigation strategies and select the right approach for a given constraint
- 6 Conduct a bias audit using IBM AI Fairness 360 and interpret the output metrics

Bias vs Hallucination — A Critical Distinction

AI Bias

A systematic tendency in an AI system's outputs that produces unfair, inequitable, or discriminatory results for specific groups — independent of whether those outputs are factually accurate. Bias is about fairness; hallucination is about accuracy. A model can be biased without hallucinating, and hallucinate without being biased.

- Hallucination: model states an incorrect fact ('the Eiffel Tower is in London') — a truth/accuracy failure
- Bias: model recommends men over equally qualified women for engineering roles — a fairness/equity failure
- Critical: anti-discrimination law applies to biased AI outputs even if technically 'accurate' by the model's training objective
- Mitigation diverges: hallucinations → RAG and grounding; bias → fairness audits, resampling, and debiasing
- Both can coexist: a medical AI may hallucinate about a rare disease AND produce biased treatment recommendations across racial groups

Three Sources of Bias: Data → Algorithm → Deployment

DATA BIAS

- 1 Historical bias: reflects past discrimination in training data
- 2 Representation bias: under/over representation of groups
- 3 Measurement bias: different error rates across groups

ALGORITHM BIAS

- 1 Model amplifies existing data bias during training
- 2 Feature selection encodes proxy variables for protected attrs
- 3 Optimisation objective ignores group-level fairness

DEPLOYMENT BIAS

- 1 Use in contexts different from training distribution
- 2 Feedback loops reinforce initial biased decisions
- 3 Unequal access to model or appeals mechanism

Bias introduced at any stage propagates through the pipeline — detection must happen at every stage, not just at training time.

Five Types of Bias in AI Systems

Bias Type	Description	Example	Detection Method
Historical	Training data reflects past human discrimination	Hiring model trained on male-dominated tech resumes	Compare model outputs to pre-bias reference distribution
Representation	Groups unequally represented in training data	Facial recognition trained on 80% white faces	Measure demographic proportions in dataset
Measurement	Different error rates in data collection across groups	Pulse oximeter calibrated on lighter skin	Compare label quality across demographic subgroups
Aggregation	Single model for heterogeneous groups	Diabetes risk model ignoring ethnic metabolic differences	Test performance stratified by subgroup
Evaluation	Metric does not capture group-level disparities	Accuracy 95% overall but 72% for minority group	Measure per-group accuracy, FPR, FNR separately

Disparate Impact and the 4/5ths Rule

Disparate Impact (US EEOC Uniform Guidelines, 1978)

A legal and statistical doctrine stating that a selection practice has an adverse disparate impact on a protected group if the selection rate for that group is less than 4/5ths (80%) of the selection rate for the group with the highest rate. Originally developed for hiring; now applied to AI-driven decisions.

- Statistical parity (demographic parity): $P(\hat{Y}=1 \mid A=0) = P(\hat{Y}=1 \mid A=1)$ — the positive outcome rate must be equal across groups
- 4/5ths rule: adverse impact if $\text{selection_rate_minority} / \text{selection_rate_majority} < 0.80$
- Legal significance: disparate impact does not require intent to discriminate — statistical evidence is sufficient for a prima facie case
- Application: hiring, lending (Equal Credit Opportunity Act), housing (Fair Housing Act), sentencing, benefits allocation
- Limitation: achieving perfect statistical parity may conflict with predictive accuracy — requires business necessity justification

Disparate Impact Analysis — Worked Calculation

$$\text{Adverse Impact Ratio} = \text{Selection Rate (Protected Group)} / \text{Selection Rate (Reference Group)}$$

1. Scenario: an AI screening tool reviewed 1,000 job applicants — 600 men and 400 women
2. Men selected: $180 / 600 = 30\%$ selection rate
3. Women selected: $72 / 400 = 18\%$ selection rate
4. Adverse Impact Ratio = $18\% / 30\% = 0.60$
5. Interpretation: $0.60 < 0.80$ (4/5ths threshold) → ADVERSE IMPACT DETECTED
6. Regulatory result: the employer bears the burden to demonstrate the selection procedure is job-related and consistent with business necessity

Note: The 4/5ths rule is a rule of thumb, not a hard legal standard — courts examine statistical significance (Fisher's exact test, chi-squared) alongside the ratio.

Individual Fairness vs Counterfactual Fairness

Individual Fairness

- Principle: similar individuals should receive similar predictions, regardless of group membership
- Metric: Lipschitz condition — $|f(x_1) - f(x_2)| \leq d(x_1, x_2)$ for a task-appropriate distance metric d
- Advantage: does not require defining protected group membership
- Challenge: defining a 'fair' distance metric d is non-trivial and domain-specific
- Example: two loan applicants with identical credit history but different race should receive identical credit decisions
- Tool: Fairlearn's individual fairness utilities

Counterfactual Fairness

- Principle: a decision is fair if it would remain the same in a counterfactual world where only the protected attribute changed
- Test: would the outcome change if the applicant's race/gender/age were different, all else equal?
- Formal: $\hat{Y}(A=a) = \hat{Y}(A=a')$ for all individuals and attribute values a, a'
- Advantage: captures causal fairness, not just correlational fairness
- Challenge: requires a causal model of the data generating process
- Example: replace 'Michael' with 'Jamal' in a resume — does the AI score change? If yes → bias

Bias Audit Process — Six-Step Workflow

1 Define Scope

Identify protected attributes (race, gender, age, disability). Define decision being audited. Specify legal standard.

2 Collect Stratified Data

Obtain representative test set with demographic labels. Verify label quality is equal across groups.

3 Compute Fairness Metrics

Calculate statistical parity, equalized odds, calibration. Compare against threshold (4/5ths, $\pm 10\text{pp}$).

4 Diagnose Root Cause

Trace bias to data, features, or model. Use SHAP to identify which features encode protected attributes.

5 Apply Mitigation

Choose pre-, in-, or post-processing strategy. Re-measure all fairness metrics after mitigation.

6 Document & Monitor

Publish audit report. Set up continuous monitoring. Schedule re-audit quarterly or on model update.

Bias Detection Tools — Overview

Tool	Provider	Type	Key Metrics	Language
AI Fairness 360 (AIF360)	IBM (OSS)	Detection + Mitigation	Statistical parity, equalized odds, disparate impact, calibration	Python
What-If Tool	Google	Interactive Visual	Counterfactual analysis, partial dependence, performance by slice	Python / TF
Fairlearn	Microsoft (OSS)	Detection + Mitigation	Demographic parity, equalized odds, bounded group loss	Python / sklearn
Aequitas	CMU (OSS)	Audit-focused	False discovery rate parity, false negative rate parity, disparity ratios	Python
Themis-ML	OSS	Statistical Testing	Feldman disparate impact remover, relabelling	Python

Pre-Processing Mitigation Strategies

Pre-Processing Mitigation

Techniques applied to training data before model training to reduce bias at the source. Pre-processing does not require access to the model architecture — it operates purely on the dataset.

- Resampling: oversample under-represented groups or undersample over-represented groups to achieve demographic balance
- Reweighting: assign higher sample weights to examples from disadvantaged groups so the model pays more attention to them during training
- Data augmentation: generate synthetic examples for under-represented groups using methods like SMOTE or GAN-based augmentation
- Label correction: identify and correct mislabelled examples that disproportionately affect specific groups
- Privilege suppression: remove or obscure features that directly or indirectly encode protected attributes (but beware of proxy features)

- Best for: when you have access to training data and can re-train the model; can be combined with in/post-

In-Processing vs Post-Processing Mitigation

In-Processing (During Training)

- Add fairness constraints to the loss function during training
- Adversarial debiasing: simultaneously train a classifier and an adversary that predicts the protected attribute — the classifier learns to fool the adversary
- Fairness regularisation: add $\lambda \cdot \text{fairness_penalty}$ to the standard cross-entropy loss
- Advantage: model learns fair representations — fairness is baked in
- Requires access to model training code and re-training budget
- Tools: AIF360 AdversarialDebiasing, Fairlearn reductions approach

Post-Processing (After Training)

- Adjust model output thresholds per group to equalise error rates
- Calibrated Equal Odds: find group-specific thresholds that achieve equal TPR and FPR
- Reject option classification: abstain from prediction in the uncertain boundary region where bias is highest
- Advantage: works with any black-box model — no access to training required
- May reduce overall accuracy — there is always an accuracy-fairness tradeoff
- Tools: AIF360 CalibratedEqualizedOdds, Fairlearn ThresholdOptimizer

Case Study — Amazon Hiring AI Gender Bias (2014–2018)

Case: Amazon's ML-Based CV Screening Tool

- Amazon built a ML hiring tool trained on 10 years of CV submissions — predominantly male in technical roles
- The model learned to penalise CVs containing the word 'women's' (e.g. 'women's chess club', 'women's engineering society')
- It also downgraded graduates of all-women's colleges and applicants with typically female names
- Amazon's team discovered the bias in 2015 and attempted fixes — but the model kept finding new proxy features encoding gender
- Amazon scrapped the tool in 2018, deciding it could not be made fair after three years of remediation attempts

Key Lesson

Root causes: (1) Historical bias in training data — the model learned that successful hires were historically male; (2) Proxy features — 'women's' as a token encoded gender even after explicit gender features were removed; (3) Evaluation failure — accuracy on historical data was high, masking the discrimination. Lesson: achieving statistical parity on proxy-free features is insufficient — the model can rediscover protected attributes through correlated features.

Counterfactual Fairness Test — Worked Example

Testing whether a hiring AI treats identical candidates differently based on name alone

Resume A (Identical qualifications — 'traditional' name)

Name: Michael Johnson
Education: B.Tech Computer Science, IIT Delhi, CGPA 8.7
Experience: 3 years Python/ML at mid-size company
AI Screening Score: 87 / 100 → Forwarded to HR

Resume B (Identical qualifications — name swap only)

Name: Mohammed Hassan [only change: first and last name]
Education: B.Tech Computer Science, IIT Delhi, CGPA 8.7
Experience: 3 years Python/ML at mid-size company
AI Screening Score: 73 / 100 → Rejected

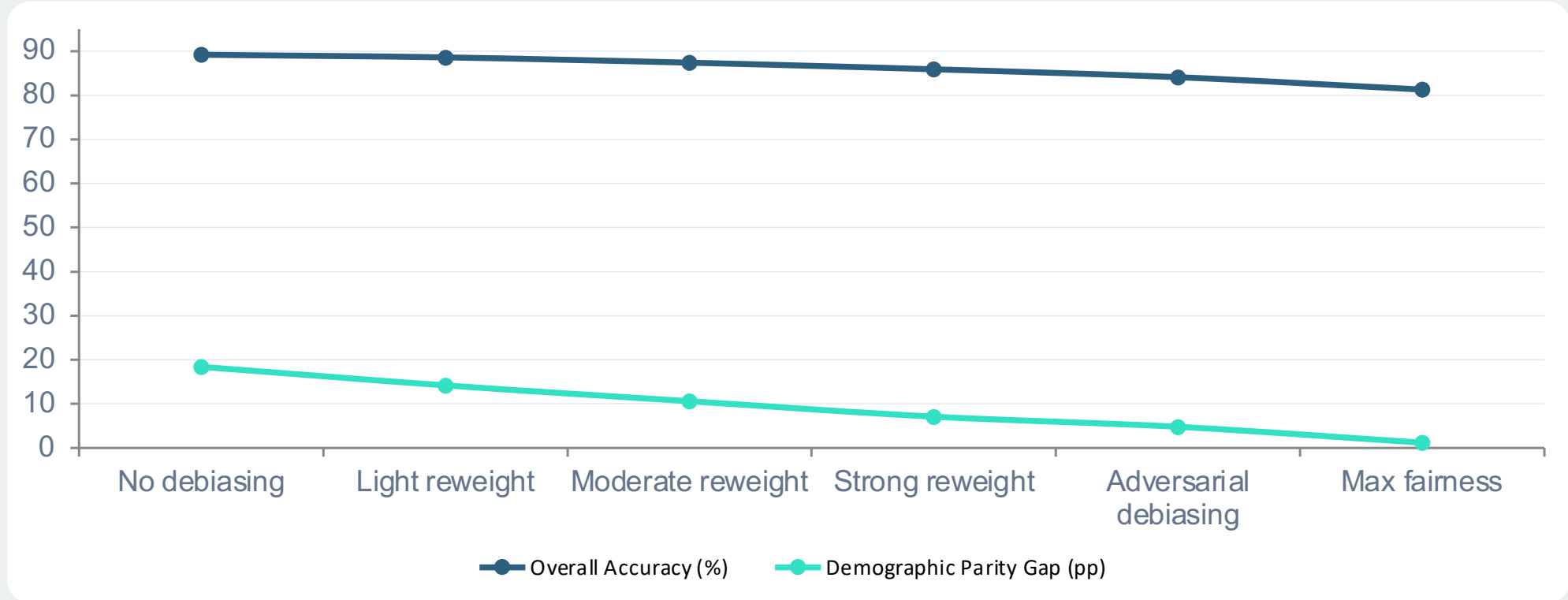
Counterfactual test result: 14-point score gap on identical qualifications. The only difference is the name — a proxy for ethnicity and religion. This is direct evidence of biased model behaviour requiring immediate investigation and remediation (post-processing threshold adjustment + training data audit).

Six Principles for Bias-Aware AI Development

Embedding bias awareness throughout the AI development lifecycle:

- Audit before deploy: no AI system affecting consequential decisions (hiring, lending, healthcare, justice) should go to production without a formal bias audit covering all relevant protected attributes
- Measure fairness alongside accuracy: add fairness metrics (demographic parity gap, equalized odds gap) to every model evaluation dashboard — not just accuracy, F1, or AUC
- Document the training data: publish a datasheet for every training dataset listing demographic composition, known biases, and data provenance — this enables downstream teams to assess fitness-for-purpose
- Test for proxy features: removing protected attributes from features is necessary but not sufficient — identify and evaluate all features correlated with protected attributes (ZIP code → race; name → gender/ethnicity)
- Human review for high-stakes: AI decisions on hiring, credit, parole, and benefits must have a human review pathway — automated-only decisions in high-stakes domains are prohibited under GDPR Article 22
- Monitor for bias drift: model fairness can degrade over time as user populations shift — schedule quarterly fairness re-audits and alert on demographic parity gap widening by > 5pp

Accuracy-Fairness Tradeoff Curve



Source: illustrative from AIF360 benchmark (Credit dataset). Every 1pp reduction in demographic parity gap costs ~ 0.4 pp accuracy. Select the operating point based on the business and legal context — there is no universally correct balance.

Key Takeaways

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- 1 Bias is a fairness failure; hallucination is an accuracy failure — they require different mitigation strategies and both can coexist in the same model
- 2 Bias enters at three stages: data collection, model training, and deployment — detection and mitigation must be applied at all three stages, not just during training
- 3 The 4/5ths rule (adverse impact ratio < 0.80) is the regulatory standard for disparate impact in consequential AI decisions — compute it for every protected attribute before deployment
- 4 Removing protected attributes from features is necessary but not sufficient — proxy features (name, ZIP code, language style) can re-introduce bias through correlation
- 5 Every major mitigation approach (pre/in/post-processing) involves an accuracy-fairness tradeoff; document the tradeoff explicitly and justify the operating point for the deployment context